CHANDU PRASAD BHAIRAPU



AI Engineer • Agentic AI, AI Coding Agents & LLM Orchestration

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ABOUT ME

AI engineer with an MSc in Artificial Intelligence and 3 years of professional experience in AI/ML engineering, most recently building agentic AI systems. I build with AI coding agents (Claude Code with custom skills) and build the parts they depend on: LLM orchestration with LangGraph and LangChain, MCP services, evaluation frameworks that measure output against ground truth, and delivery with Docker, GitLab CI/CD and AWS CDK, plus apps deployed on Azure. I turn domain experts' requirements into systems they can check and approve themselves.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an agentic knowledge-extraction pipeline end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts from the German energy sector to turn their requirements into a working system.
- Designed LLM orchestration for multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) and evaluated it against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Built custom Claude Code skills and agentic coding workflows for my own development, keeping agent-written code to clean code principles; paired with the MCP service, the agent works directly against the knowledge graph.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream instead of an LLM approach, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, Claude Code, MCP, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, Neo4j, MongoDB, Postgres, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes decisions across vector database queries, general inference, and summarization.
- Built a Python backend with FastAPI and a React front end, with API testing in Postman.
- Containerized the stack with Docker and deployed it to AWS using CDK (infrastructure as code); version-controlled in GitLab with a CI/CD pipeline.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Built an automated evaluation framework for RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.

- **Tech stack: Python, LangChain, RAGAS, PytrecEval, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, benchmarked against empirical curve-fitting.
- Built a Streamlit dashboard for chemists to simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Pandas, Scikit-Learn, XGBoost, TensorFlow/Keras, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy on unseen data.

PROJECTS

Project: Spec-Driven Coding Agent Harness | Claude Code | MCP | Langfuse | GitHub Actions

- Harness that turns a plain-language feature request into testable specifications (acceptance criteria written as pytest tests) and lets Claude Code implement them under a governance rule set across the SDLC: project conventions, custom skills, and hooks that block edits to protected files and secrets; an MCP server gives the agent the spec and task context.
- A GitHub Actions CI/CD pipeline gates every agent-written change with automated testing, static security analysis, and dependency scanning (DevSecOps), plus an agent evaluation suite over benchmark tasks (pass rate, cost, turns per task); traces of the agent loops go to Langfuse for observability.

TECHNICAL SKILLSET

- **AI Coding Agents & Agentic AI:** Claude Code (custom skills, agentic coding workflows), MCP, LangGraph (multi-agent orchestration), LangChain, LiteLLM, prompt engineering
- **LLM Applications & Evaluation:** RAG, GraphRAG, evaluation frameworks (RAGAS, PytrecEval), Hugging Face Transformers, embedding benchmarking
- **Software Delivery & Cloud:** CI/CD (GitLab CI/CD), Docker, Kubernetes, AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3), Azure, API testing (Postman)
- **Programming & APIs:** Python, SQL, FastAPI, REST APIs, Git, React (beginner), Streamlit
- **Databases:** Postgres, MongoDB, Neo4j, SPARQL, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis
- **Ways of working:** Agile

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)